2D Matrices: Transformations

Questions 1–10

Question 1. A transformation T is represented by the matrix $\mathbf{A} = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$

a) State the geometrical transformation that T represents.
b) Find the eigenvalues of matrix $\mathbf{A}$ and their corresponding eigenvectors.
c) State any invariant lines of the transformation.
d) Find $\det \mathbf{A}$ and give a geometrical interpretation of this value.

Question 2. A transformation T is represented by the matrix $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

a) State the geometrical transformation that T represents.
b) Find the eigenvalues of matrix $\mathbf{A}$ and their corresponding eigenvectors.
c) State any invariant lines of the transformation.
d) Find $\det \mathbf{A}$ and give a geometrical interpretation of this value.

Question 3. A transformation T is represented by the matrix $\mathbf{A} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$

a) State the geometrical transformation that T represents.
b) Find the eigenvalues of $\mathbf{A}$
c) By considering the eigenvalues and then geometrically, explain why there are not any invariant lines of the transformation.
d) Find $\det \mathbf{A}$ and give a geometrical interpretation of this value.

Question 4. A transformation T is represented by the matrix $\mathbf{A} = \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}$

a) Find the eigenvalues of matrix $\mathbf{A}$ and their corresponding eigenvectors.
b) State any invariant lines of the transformation.
c) Find $\det \mathbf{A}$ and give a geometrical interpretation of this value.
d) Describe fully the transformation represented by $\mathbf{A}$

Question 5. A transformation T is represented by the matrix $\mathbf{A} = \frac{1}{5}\begin{pmatrix} 13 & 6 \\ 6 & 22 \end{pmatrix}$

a) Find the eigenvalues of $\mathbf{A}$
b) Determine the Cartesian equations of the two lines passing through the origin which are invariant under the transformation T
c) Describe fully the transformation represented by $\mathbf{A}$

Question 6. Under a linear transformation, T, the point $(3, 4)$ is mapped to the point $(6, 8)$
For the matrix representing T:

a) Write down the eigenvector.
b) Write down the corresponding eigenvalue.

Question 7. Under a linear transformation, T, the point $(2, -5)$ is mapped to the point $(-4, 10)$
For the matrix representing T:

a) Write down the eigenvector.

b) Write down the corresponding eigenvalue.

Question 8. A transformation T is represented by the matrix $\mathbf{A} = \begin{pmatrix} p & q \\ q & p \end{pmatrix}$ for non-zero constants p and q where $p \neq q$

a) Find the eigenvalues and corresponding eigenvectors of matrix $\mathbf{A}$ giving your answer in terms of p and q

b) Show that the invariant lines of the transformation are perpendicular.

c) Describe fully the transformation represented by $\mathbf{A}$

Question 9. A transformation T is represented by the matrix $\mathbf{A} = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$

a) Find the eigenvalues of $\mathbf{A}$
b) Determine the Cartesian equations of the two lines passing through the origin which are invariant under the transformation T
c) Describe fully the transformation represented by $\mathbf{A}^2$

Question 10. A transformation T is represented by the matrix $\mathbf{A} = \begin{pmatrix} -10 & 8 \\ -16 & 14 \end{pmatrix}$

Describe fully the transformation represented by $\mathbf{A}^3$